3.4 Function Notation

Algebra I 3.4 · Student sheet

Name _____ Period ____ Date _____

I can…

- evaluate functions using function notation.
- find the input that gives a certain output.
- graph linear functions and interpret function notation in real life.

Vocabulary · fill in the missing word

Function notation — f(x), read "f of x," is the _____ of function f for the input x

f(a) = b — the point (a, b) is on the _____ of f

Part 1 · Function machine

f(x) = 2x + 5 g(x) = x² − 1

Input x	f(x)	g(x)

Output f(x)	Input x
15	
−3	

Part 2 · Notes and examples (fill in each note, then work the example)

Function notation: f(x) is read "_____."

Evaluating: To evaluate f(3), _____ 3 for x.

Example 1

Let f(x) = −4x + 7. Find f(2) and f(−3).

Solving: To find x when f(x) = 15, set the rule equal to 15 and _____.

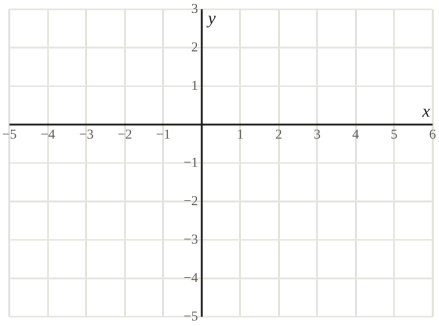
Example 2

Let g(x) = 5x − 3. Find x so that g(x) = 22.

On the graph: f(3) = 11 means the point _____ is on the graph.

Example 3

Graph $f(x) = \frac{1}{2}x - 2$. Use the graph to find f(4) and the x-value where f(x) = −3.



Example 4

The cost (in dollars) of renting a truck and driving m miles is C(m) = 0.25m + 15. Find C(100) and explain what it means.

Part 3 · Partner practice

Partner 1

Let $h(x) = x^2 + 1$. Find h(−3).

Partner 2

Let f(x) = 3x + 4. Find x so that f(x) = −8.